

Rusham R. Bhatt

US Citizen | rbhatt4@umbc.edu | [LinkedIn](#) | [GitHub](#)

EDUCATION

University of Maryland, Baltimore County (UMBC)

B.S. in Computer Science and Minor in Computational Biology and Bioinformatics

Meyerhoff Scholar | Member of the Honors College

May 2026

GPA: 3.84/4.00

EXPERIENCE

Software Development Intern

Jun 2025 - Aug 2025

Salk Institute for Biological Studies (PI: T. Pereira, PhD)

San Diego, CA

- Developed a 3D GUI to map animal motion capture data onto virtual animal skeletal models using PyGFX and MuJoCo, collaborating with team members to align with user requirements.
- Enhanced pre-processing workflows to boost registration speed and improve modeling accuracy, contributing to iterative code reviews and debugging sessions.

Undergraduate ML Researcher

Feb 2025 - May 2025

Vinjamuri Lab (PI: R. K. Vinjamuri, PhD)

Baltimore, MD

- Engineered a neural network using Python to classify left/right hand movement from EEG signals.
- Pre-processed neural data to enhance signal clarity and boost classification performance, documenting methodologies and collaborating with peers.

Bioinformatics Research Intern

May 2024 - Aug 2024

Harvard Medical School (PI: M. Farhat, MD)

Boston, MA

- Developed a standardized Python pipeline for analyzing genetic variants in M. tuberculosis WGS data, demonstrating proficiency in Python and software development.
- Streamlined genomics processing for lab-wide deployment by improving pipeline efficiency, actively participating in code reviews and debugging efforts.

Software Development Intern

May 2023 - Aug 2023

RCSB Protein Data Bank, Rutgers University (PI: S. K. Burley, MD, PhD)

New Brunswick, NJ

- Developed a Python package to facilitate easy use of the RCSB PDB search API, enabling researchers to retrieve structural biology data directly from within a Python environment.
- Engaged in code reviews and debugging to ensure code quality.

PRESENTATIONS

Leadership Alliance National Symposium (LANS) [Poster]

July 2024

Harvard Summer Honors Undergraduate Research Program (SHURP) Final Symposium [Oral]

July 2024

Annual Biomedical Research Conference for Minoritized Students (ABRCMS) [Poster]

Nov 2023

- Travel award recipient.

Rutgers University Research Intensive Summer Experience (RISE) Final Symposium [Oral and Poster]

July 2023

PROJECTS

Multimodal Sensor-Text Representation Alignment (Advisor: Dr. KMA Solaiman)

2025

- Encoding sensor/time signal data and corresponding text labels in latent space to determine similarities or correlations between sensor and text embeddings.
- Determined embeddings by processing data through encoder models such as SentenceBERT and TS2Vec.

College Interactive 4-year Plan

2025

- Built a drag-and-drop planner with a team of 4 students using HTML, CSS, and JavaScript for UMBC students.
- Designed and deployed the backend with Node.js and MongoDB to store and manage course data.
- Applied the Scrum framework, delivering features iteratively in sprints to ensure continuous progress and collaboration.

2D Mini Dungeon Game

2024

- Built a C++ game using SDL2; implemented OOP design and real-time rendering.

SKILLS

Programming: Python, C++, Shell Scripting, C, JavaScript, HTML/CSS

Tools: Jupyter, Git, PyPi, Pandas, Matplotlib, Snakemake, ReactJS, NodeJS, MongoDB, PyGFX, MuJoCo, Qt, PyTorch, Tensorflow, Numpy

PUBLICATIONS/MANUSCRIPTS

Piehl, D. W., Vallat, B., Truong, I., Morsy, H., **Bhatt, R.**, Blaumann, S., ... & Burley, S. K. (2025). rcsb-api: Python Toolkit for Streamlining Access to RCSB Protein Data Bank APIs. *Journal of molecular biology*, 168970.

Role: Contributed core functionality of the Python search API package.

Zhang, C. Y., Yang, Y., Sirbu, A., Abe, E. T., Wärnberg, E., Leonadis, E. J., ...**Bhatt, R.**,...& Pereira, T. D. (2025). MIMIC-MJX: Neuromechanical Emulation of Animal Behavior. *arXiv preprint arXiv:2511.20532*.

Role: Developed 3D GUI for mapping keypoints between motion capture data and virtual MuJoCo animal models.

ACTIVITIES AND LEADERSHIP

Bio Quantitative Methods Workshop Participant

Jan 2024

Massachusetts Institute of Technology

Cambridge, MA

- Participated in daily lectures, hands-on computer labs in which participants learn to apply quantitative tools and Python to experimental data analysis.

CMSC 201 Teaching Assistant

Aug 2023 - Present

UMBC College of Engineering and Information Technology

Baltimore, MD

- Teaching assistant for gateway computer science course focused on Python.
- Responsible for grading lab assignments, homeworks, projects, and exams.
- Lead lab discussions, present slides on Python programming, answer questions, and help students understand and write proper Python code.
- Proctor midterm and final exams.

STEM Tutor

Oct 2022 - Jan 2024

STEMCX

Baltimore, MD

- Tutored Baltimore City secondary school students on various STEM subjects once or twice a week.